

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Concept Mapping
Weightage:	35%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	B / 2nd year	Date:	17/07/26

Instructions to Students

- Identify the key computer architecture concepts, components, and performance parameters mentioned in the given topic or scenario.
- Organize the identified concepts logically by establishing relationships between CPU, memory, I/O devices, bus structures, instruction execution cycle, and performance metrics.
- Represent the concepts using an appropriate concept map with clear nodes, connecting lines, and meaningful linking phrases.
- Demonstrate the hierarchy and interdependence among concepts, showing how one concept influences or interacts with another.
- Ensure the concept map is complete, accurate, well-structured, and easy to understand by using appropriate terminology and logical connections.

QUESTIONS

- A video editing application is exporting a 4K movie, where the CPU continuously fetches video frames and audio data from an SSD, applies filters and compression, and writes the processed output to a display preview and a storage device. Construct a concept map linking the various units of the computer, using labelled arrows to illustrate the flow of instructions and data. How do the five functional units coordinate to retrieve multimedia data, process video and audio operations, and produce the final output during video rendering?
- A software engineer notices that a data analytics application is taking longer than expected to process large datasets. To identify the cause, the engineer analyses how each instruction progresses through the Fetch, Decode, and Execute stages and examines performance factors such as CPI, Clock Cycles, Clock Rate, and CPU Execution Time. Construct a concept map linking the Instruction Execution Cycle with these performance parameters. How do the Fetch-Decode-Execute stages influence CPU execution time, and how can the performance factors be used to evaluate and improve processor performance?
- A robotics engineer is selecting a microprocessor for an autonomous mobile robot and must compare the 8085 and 8086 architectures to determine which offers better execution speed and memory management. Construct a concept map comparing the 8085 and 8086 by linking their instruction sets, register organization, addressing modes, memory organization, and memory segmentation (8086). Show how these architectural features influence program execution, memory access, multitasking capability, and efficiency in real-time applications.
- A software developer is optimizing an embedded control program and must choose the most suitable addressing mode for different operations to achieve faster execution and efficient memory usage. Construct a concept map linking Immediate, Direct, Register, Indirect, and Indexed Addressing Modes with Data Transfer, Arithmetic, and Control Instructions. Show how each addressing mode influences memory access time, instruction size, execution speed, and processor efficiency, and include real-world applications where each addressing mode is most appropriate.
- A firmware engineer is debugging an embedded system where machine instructions are stored in memory in binary but displayed in hexadecimal for easier analysis. Construct a concept map linking Binary and Hexadecimal Number Systems with instruction representation, memory storage, CPU instruction interpretation, data conversion, and debugging. Show how data is converted between binary and hexadecimal, how instructions are stored and executed by the CPU, and why hexadecimal notation is preferred in low-level programming and system design. How do binary and hexadecimal number systems support instruction representation, memory organization, CPU execution, and debugging in computer systems?

Code of Conduct

I U. Vishwa / 19251192 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: U. Vishwa

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

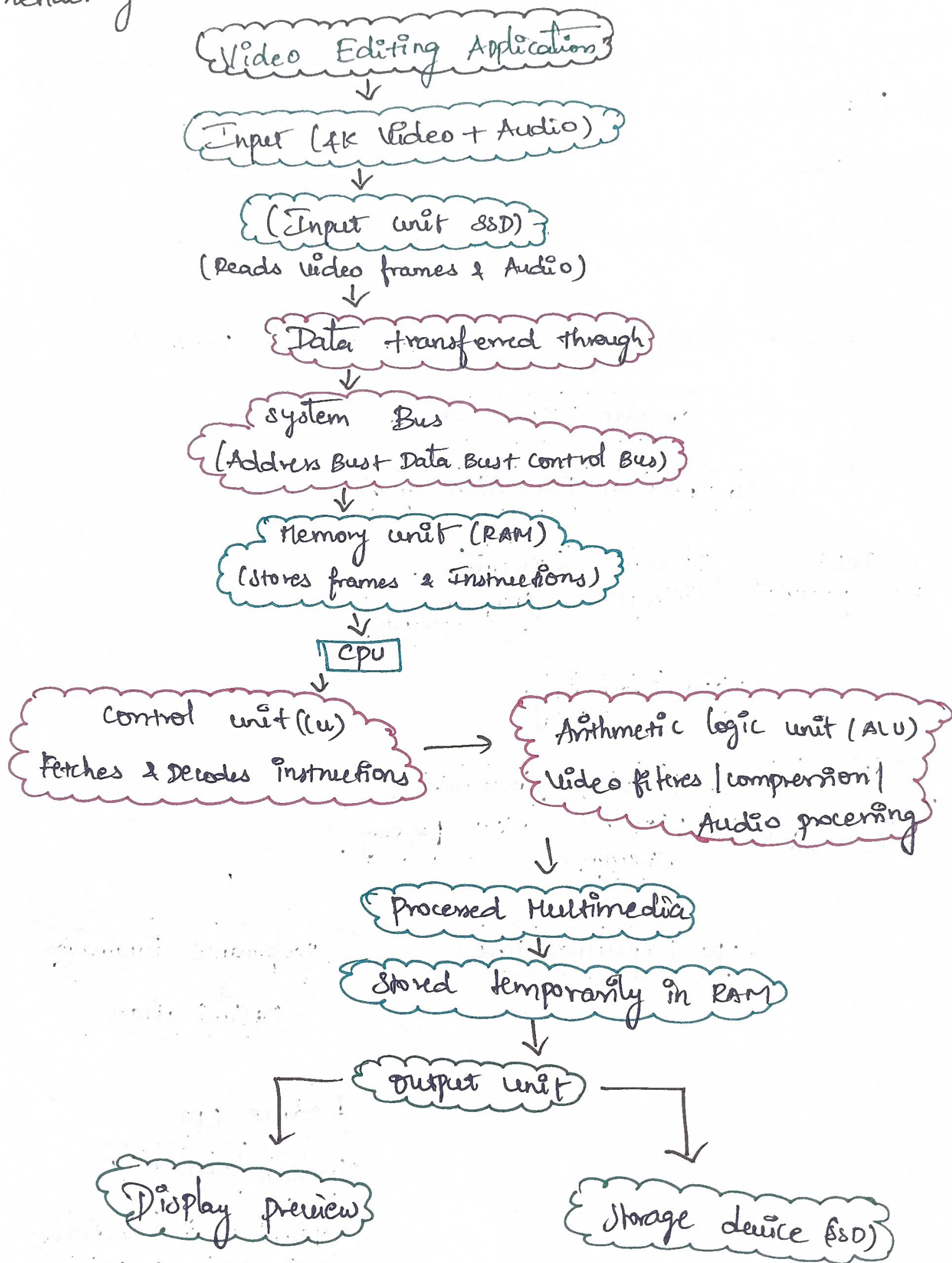
Marks Evaluation:

Q. No	Concept Identification (25)	Linkages (25)	Organization (20)	Integration of Literature (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

Course Coordinator

Faculty Incharge

Q1. Functional units of computer system During video rendering.



Functional unit coordination:

- * Input unit - Reads multimedia data from SD.
- * Memory unit - stores instructions and temporary frames
- * CPU (CU + ALU) - Executes filters, compression, encoding
- * Bus system - Transfers instructions and data
- * Output unit - Displays preview and stores rendered video.

Q2 Instruction Execution cycle & performance metrics:

Program Execution

Instruction Execution cycle

Fetch

(read instruction)

Decode

(interpret)

Execute

(performance operation)

cpu processing

clock cycles

(number of cycles)

cpi

(cycles/
Instructions)

clock rate

(GHz frequency)

cpu Execution time

performance Evaluation

&
Optimization

Reduce cpi

Increase clock rate

Optimize instructions.

Formula:

$$\text{CPU Execution time} = \frac{\text{Instruction count} \times \text{CPI} \times \text{clock cycle time}}{\text{clock rate}}$$

OR

$$\text{CPU Execution time} = \frac{(\text{Instruction count} \times \text{CPI})}{\text{clock rate}}$$

Influence:

- Fetch delays increases clock cycles.
- Complex decoding increases CPI.
- Faster execution reduces CPU time.
- Higher clock rate improves performance.

Q3 comparison of 8085 and 8086 Architecture:

8085

8-bit processor

64KB memory

No segmentation

code data

Register organization

8086

16-bit processor

1MB memory

Memory segmentation

Stack segment

Addressing

Accumulator B/C/D/E/H/L

immediate

Direct

Register

Indexed

Base

Relative

Faster program Execution

Better memory management

Improved Multitasking

Real-time Application Efficiency

Comparison table :

Feature :

	8085	8086
Word size	8-bit	16-bit
Address Bus	16-bit	20-bit
memory	64 KB	1 MB
segmentation	No	yes
Registers	limited	more registers
speed	lower	Higher
Multitasking	poor	Better

Q4 Addressing Modes:



Real world Examples:

Addressing mode

Immediate

Direct

Register

indirect

Application:

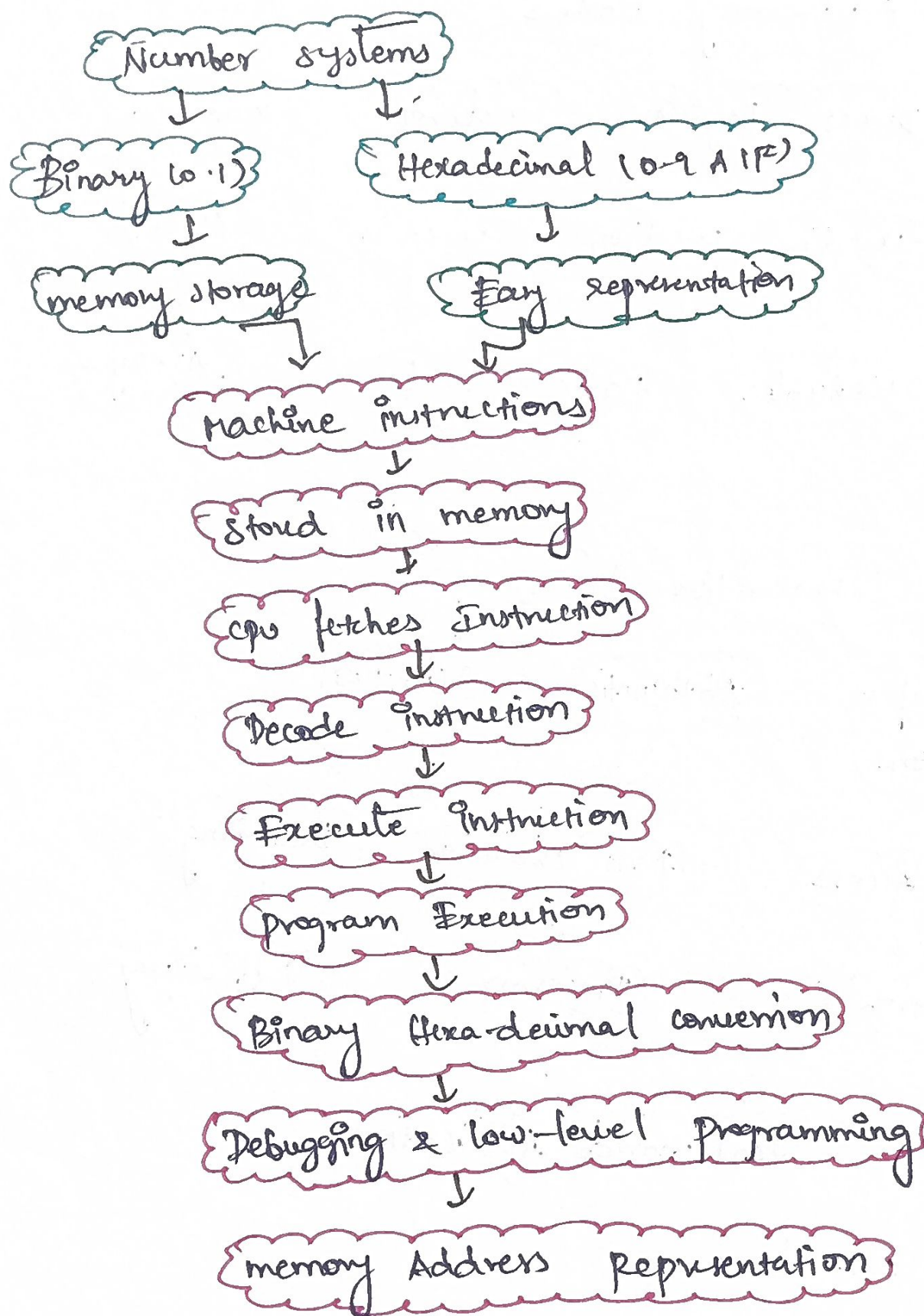
Initialize variables

Access fixed memory

Arithmetic

linked lists

Q5 Binary & Hexadecimal number systems.



Importance:

- * Binary is used internally by the CPU.
- * Hexadecimal simplifies long binary values.
- * Memory addresses are easier to read in hexadecimal.
- * Assembly programmers commonly use hexadecimal.